Hongming Qiu (Henry)

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Education

University of Illinois Urbana-Champaign

M.S. in Electrical and Computer Engineering (Thesis) — *Expected May 2028*

B.S. in Computer Engineering | Overall GPA: 3.99/4.00 | Highest Honors | Bronze Tablet — *August 2026*

Research Experience

Kernel Security Researcher — **September 2025 – Present**

UIUC | Advisor: Prof. Tianyin Xu | In collaboration with IBM Research [code] [LPC '26 talk]

- Designed and implemented **Linux kernel IMA extensions** for policy-controlled eBPF loading, measurement, and appraisal, with PKCS#7 signature verification and TPM-backed attestation.
- Built **RCU-based, race-safe reappraisal and revocation** of resident eBPF programs; removed references across links, pins, file descriptors, tail-call maps, and cgroup attachments.
- Integrated signed loading and recovery with **Cilium's networking datapath**; evaluated remediation across 34 purge configurations with zero residual eBPF programs; wrote the resulting manuscript as first author.
- Reproducing OS-CFI's origin-sensitive design to evaluate **eBPF-based Linux kernel control-flow integrity**.

LLVM Compiler & Toolchain Research — **February 2025 – Present**

Independent Research | Three-person team [code]

- Co-developed an **LLVM-based polymorphic compilation framework** for Windows and Linux; designed EliminateIAT to rewrite imports while preserving invoke control flow and variadic calls, and led MSVC/LLVM toolchain integration.
- Evaluated **1,160 artifacts** from seven malware families against up to 76 VirusTotal engines; measured 95.5% mean inter-build byte divergence and sub-1% mean detection for the strongest pass combinations.

Selected Systems Projects

RISC-V Operating System Kernel | *C/C++, RISC-V Assembly* — **2024; 2026 – Present**

- Built a C-based RISC-V kernel (ECE 391) with **process management, IPC, and device drivers**; ported Doom and Rogue to student-built kernels.
- Porting and redesigning the kernel in **C++23** for ECE 498OS with **multi-core support**, cooperative scheduling, CPU-local state, IPI-based wakeups, and RAII synchronization.
- Implemented kernel-backed **C++ allocation and runtime hooks**, polymorphic I/O, and typed intrusive containers for scheduler and filesystem lists.

Out-of-Order RISC-V Processor | *SystemVerilog, ECE 411, Team of 3* — **Spring 2026**

- Co-designed and verified a **2-way superscalar RV32IM processor** with explicit register renaming, split load/store queues, store forwarding, pipelined caches, and early branch recovery.
- Implemented **gshare and return-address prediction for two-stage fetch**, cutting mispredictions from about **60% to 10%**; sustained **over 1 IPC** on compression and ray-tracing benchmarks at 500 MHz within 300,000 μm^2.

Teaching & Leadership

ECE 391 Teaching Assistant / Course Assistant — **January 2025 – Present**

ECE Department, University of Illinois Urbana-Champaign

- Lead discussions and support **20+ students weekly** in office hours for a 250+ student RISC-V OS course.
- Cut **autograder runtime in half** across all test cases by removing redundant runs and caching builds.

Admin & Purple Team Lead — **September 2023 – Present**

SIGPwny, cybersecurity club at the University of Illinois Urbana-Champaign

- Built and maintained **multiple Proxmox instances** for a networking and security practice range, with **20+ VMs**, segmented networks, VPN access, and pfSense/OPNsense firewalls.
- Authored **30+ CTF challenges** for UIUCTF and SIGPwny; led numerous advanced workshops for 50+ club members.

Technical Skills

Languages: C, C++, Python, Go, Bash, x86 and RISC-V Assembly, SystemVerilog
Systems: Linux kernel, operating systems, device drivers, IPC, RCU, eBPF, LLVM
Networking & Infrastructure: Cilium, container networking, pfSense, OPNsense, VPNs, Proxmox
Certifications: OSED, OSCP+, Certified CyberDefender (CCD)